

WDDA FS 2026: Guide for Problem Set 8

2026-05-20

1 Introduction

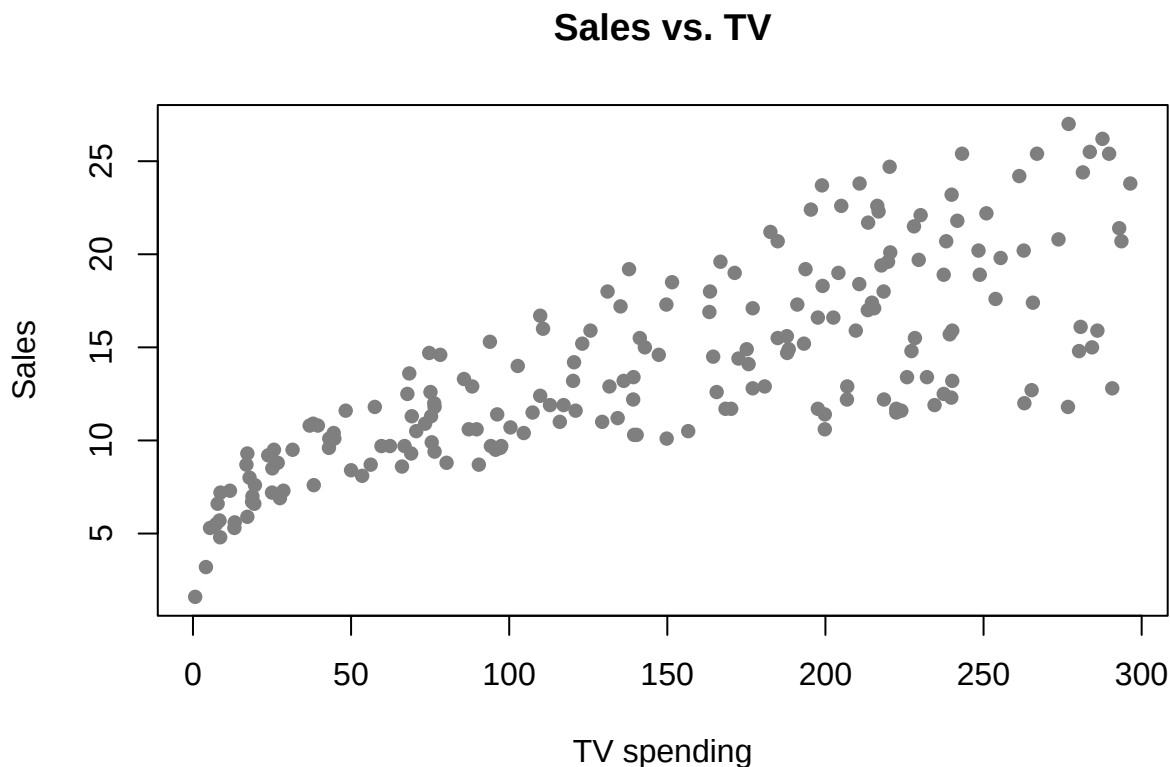
This guide provides detailed explanations for the exercises in WDDA FS 2026 Problem Set 8. The set focuses on **non-linear relationships** (variable transformations) and **categorical predictors** (dummy variables, interactions). We learn how the linear regression model can be extended to also handle non-linear and non-numeric relationships.

2 Exercise 1: Advertising – Non-linear models for TV and sales

Task: Compare different model forms for the relationship between TV spending and sales. Check whether a transformation improves model fit and whether the linear regression assumptions are better satisfied.

2.1 Step 1: Read and visualise the data (a)

```
adv <- read_excel("../data/WDDA_08.xlsx", sheet = "Advertising")
plot(adv$TV, adv$sales, pch = 16, col = "gray50",
     xlab = "TV spending", ylab = "Sales",
     main = "Sales vs. TV")
```



Assessment: The data show a clearly positive but curved relationship – the effect of additional TV spending appears to flatten at high values (diminishing marginal returns).

2.2 Step 2: Fit six models (b)

```
attach(adv)

md1 <- lm(sales ~ TV)                # Linear
md2 <- lm(sales ~ poly(TV, 2, raw = TRUE)) # Quadratic
md3 <- lm(sales ~ sqrt(TV))          # Square-root
md4 <- lm(sales ~ log(TV))           # Log(x)
md5 <- lm(log(sales) ~ log(TV))      # Log-log
md6 <- lm(log(sales) ~ TV)           # Log(y)

cat("R-squared values:\n")
```

R-squared values:

```
cat("Linear:      ", round(summary(md1)$r.squared, 4), "\n")
```

Linear: 0.6119

```
cat("Quadratic:   ", round(summary(md2)$r.squared, 4), "\n")
```

Quadratic: 0.619

```
cat("Square-root: ", round(summary(md3)$r.squared, 4), "\n")
```

Square-root: 0.6229

```
cat("Log(x):      ", round(summary(md4)$r.squared, 4), "\n")
```

Log(x): 0.565

```
cat("Log-log:     ", round(summary(md5)$r.squared, 4), "\n")
```

Log-log: 0.7421

```
cat("Log(y):      ", round(summary(md6)$r.squared, 4), "\n")
```

Log(y): 0.6156

Important warning: R^2 is only **directly comparable** between models with the same response. Models md5 and md6 (with $\log(y)$ as response) explain the variance of $\log(\text{sales})$, not of sales itself – their R^2 is therefore **not** directly comparable to the other models.

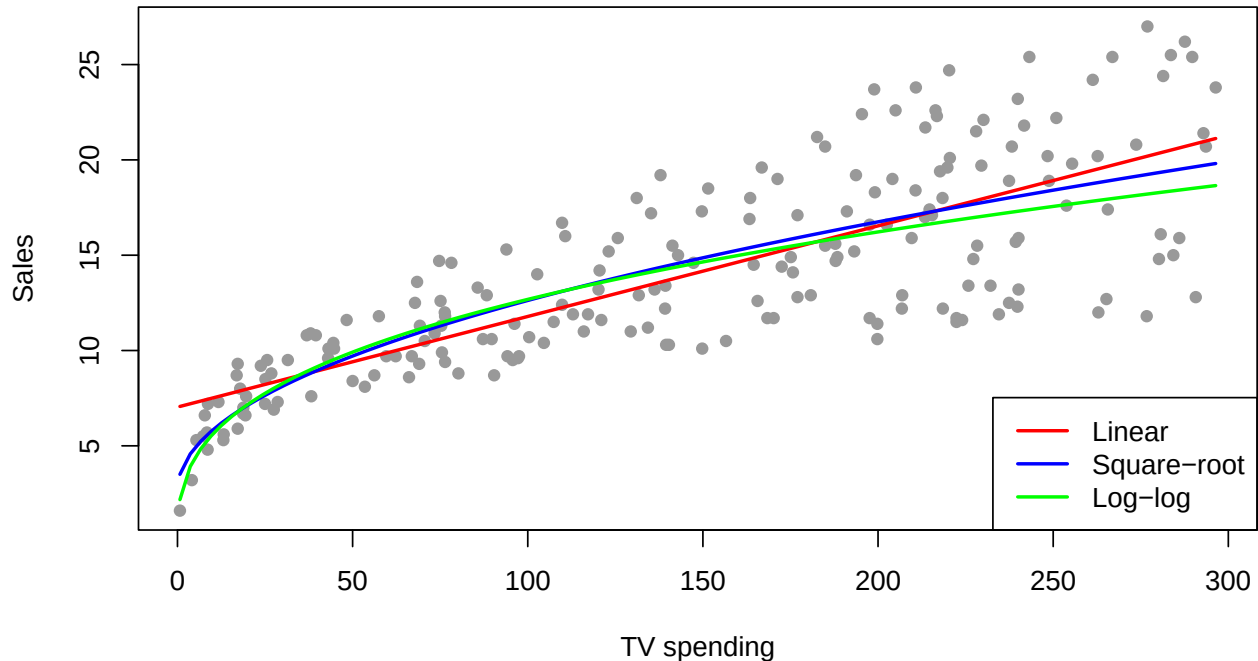
2.3 Step 3: Visualisation of the model curves (c)

```
f1 <- function(x) coef(md1)[1] + coef(md1)[2] * x
f3 <- function(x) coef(md3)[1] + coef(md3)[2] * sqrt(x)
f5 <- function(x) exp(coef(md5)[1] + coef(md5)[2] * log(x))

plot(TV, sales, pch = 16, col = "gray60",
     xlab = "TV spending", ylab = "Sales",
     main = "Comparison of regression models")
curve(f1, add = TRUE, col = "red", lwd = 2)
curve(f3, add = TRUE, col = "blue", lwd = 2)
curve(f5, add = TRUE, col = "green", lwd = 2)
```

```
legend("bottomright", legend = c("Linear", "Square-root", "Log-log"),
      col = c("red", "blue", "green"), lwd = 2)
```

Comparison of regression models

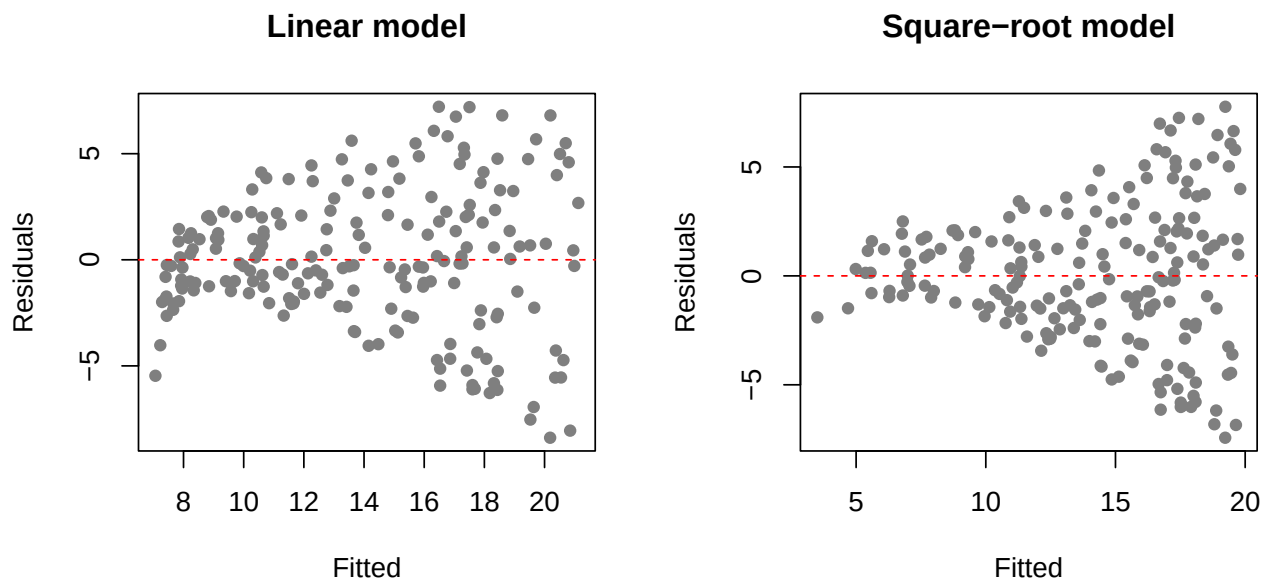


Assessment: Square-root and log-log models track the data cloud better than the linear model, especially in the upper range (high TV spending).

2.4 Step 4: Residual analysis (d)

```
par(mfrow = c(1, 2))
plot(fitted(md1), resid(md1), pch = 16, col = "gray50",
     main = "Linear model", xlab = "Fitted", ylab = "Residuals")
abline(h = 0, lty = 2, col = "red")

plot(fitted(md3), resid(md3), pch = 16, col = "gray50",
     main = "Square-root model", xlab = "Fitted", ylab = "Residuals")
abline(h = 0, lty = 2, col = "red")
```



```
par(mfrow = c(1, 1))
```

Assessment:

- For the linear model the residuals often show a **U-shaped pattern** – a hint at non-linear structure that the linear model fails to capture.
- For the square-root model the residuals are usually more evenly scattered around zero.

2.5 Step 5: Interpretation of the log-log model (e)

```
summary(md5)
```

```
##
## Call:
## lm(formula = log(sales) ~ log(TV))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.43349 -0.15917  0.01696  0.16910  0.39399
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.90525    0.07107   12.74  <2e-16 ***
## log(TV)      0.35504    0.01487   23.87  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2109 on 198 degrees of freedom
## Multiple R-squared:  0.7421, Adjusted R-squared:  0.7408
## F-statistic: 569.8 on 1 and 198 DF,  p-value: < 2.2e-16
```

Interpretation:

- The slope on $\log(\text{TV})$ is an **elasticity**: a 1% increase in TV spending leads on average to roughly $\beta_1\%$ change in sales.
- This is an economically very intuitive quantity, especially for marketing decisions.

3 Exercise 2: Default – Categorical predictors

Task: For the Default dataset, examine how account balance (`balance`) depends on student status (`student`) and income (`income`). Use dummy variables and interactions.

3.1 Step 1: Read the data (a)

```
default <- read_excel("../data/WDDA_08.xlsx", sheet = "Default")
default$student <- factor(default$student, levels = c("No", "Yes"))
head(default)
```

```
## # A tibble: 6 x 4
##   default student balance income
##   <chr>   <fct>    <dbl> <dbl>
## 1 No     Yes      1056. 16001.
## 2 No     No       1136. 38475.
## 3 No     No       994. 26769.
## 4 No     No       862. 49672.
## 5 No     No       595. 43153.
## 6 No     No       758. 55279.
```

```
table(default$student)
```

```
##
## No Yes
## 708 292
```

3.2 Step 2: Simple model with a categorical predictor (b)

```
mod_s <- lm(balance ~ student, data = default)
summary(mod_s)
```

```
##
## Call:
## lm(formula = balance ~ student, data = default)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -974.73 -379.64   -5.95   342.64 1444.62
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    776.35      18.14  42.794 < 2e-16 ***
## studentYes     198.38      33.57   5.909 4.72e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 482.7 on 998 degrees of freedom
## Multiple R-squared:  0.0338, Adjusted R-squared:  0.03284
## F-statistic: 34.92 on 1 and 998 DF, p-value: 4.716e-09
```

Interpretation:

- **Intercept:** mean balance for non-students (baseline).
- **studentYes:** difference between students and non-students.
- The p-value tests whether this difference is significant.

3.3 Step 3: Manual dummy coding (c)

```
default$st10 <- ifelse(default$student == "Yes", 1, 0)
mod_s_dummy <- lm(balance ~ st10, data = default)
```

```
cat("Factor-model coefficients:\n")
```

```
## Factor-model coefficients:
```

```
print(coef(mod_s))
```

```
## (Intercept) studentYes
```

```
##      776.3500      198.3827
```

```
cat("\nDummy-model coefficients:\n")
```

```
##
```

```
## Dummy-model coefficients:
```

```
print(coef(mod_s_dummy))
```

```
## (Intercept)      st10
```

```
##      776.3500      198.3827
```

Assessment: Both models are **identical**. The manual dummy variable is just an alternative notation.

3.4 Step 4: Multiple regression with income (d)

```
mod_si <- lm(balance ~ income + student, data = default)
summary(mod_si)
```

```
##
```

```
## Call:
```

```
## lm(formula = balance ~ income + student, data = default)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -977.73 -380.66   -3.71   345.16 1444.88
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  7.918e+02  7.158e+01  11.062  < 2e-16 ***
```

```
## income      -3.857e-04  1.729e-03  -0.223  0.823513
```

```
## studentYes   1.899e+02  5.075e+01   3.741  0.000193 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 482.9 on 997 degrees of freedom
```

```
## Multiple R-squared:  0.03385,    Adjusted R-squared:  0.03191
```

```
## F-statistic: 17.47 on 2 and 997 DF,  p-value: 3.5e-08
```

Interpretation:

- **income:** expected change in balance per unit increase in income, **at constant student status**.
- **studentYes:** difference in balance between students and non-students, **at constant income**.
- Geometrically: two **parallel** regression lines – only the level (intercept) differs.

3.5 Step 5: Model with interaction (e)

```
mod_si_int <- lm(balance ~ income * student, data = default)
summary(mod_si_int)

##
## Call:
## lm(formula = balance ~ income * student, data = default)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1007.17  -383.56   -3.95   343.47  1444.69
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.809e+02  7.397e+01  10.558  <2e-16 ***
## income        -1.139e-04  1.790e-03  -0.064   0.9493
## studentYes     2.692e+02  1.443e+02   1.865   0.0624 .
## income:studentYes -4.063e-03  6.922e-03  -0.587   0.5573
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 483.1 on 996 degrees of freedom
## Multiple R-squared:  0.03419,    Adjusted R-squared:  0.03128
## F-statistic: 11.75 on 3 and 996 DF,  p-value: 1.435e-07
```

Interpretation:

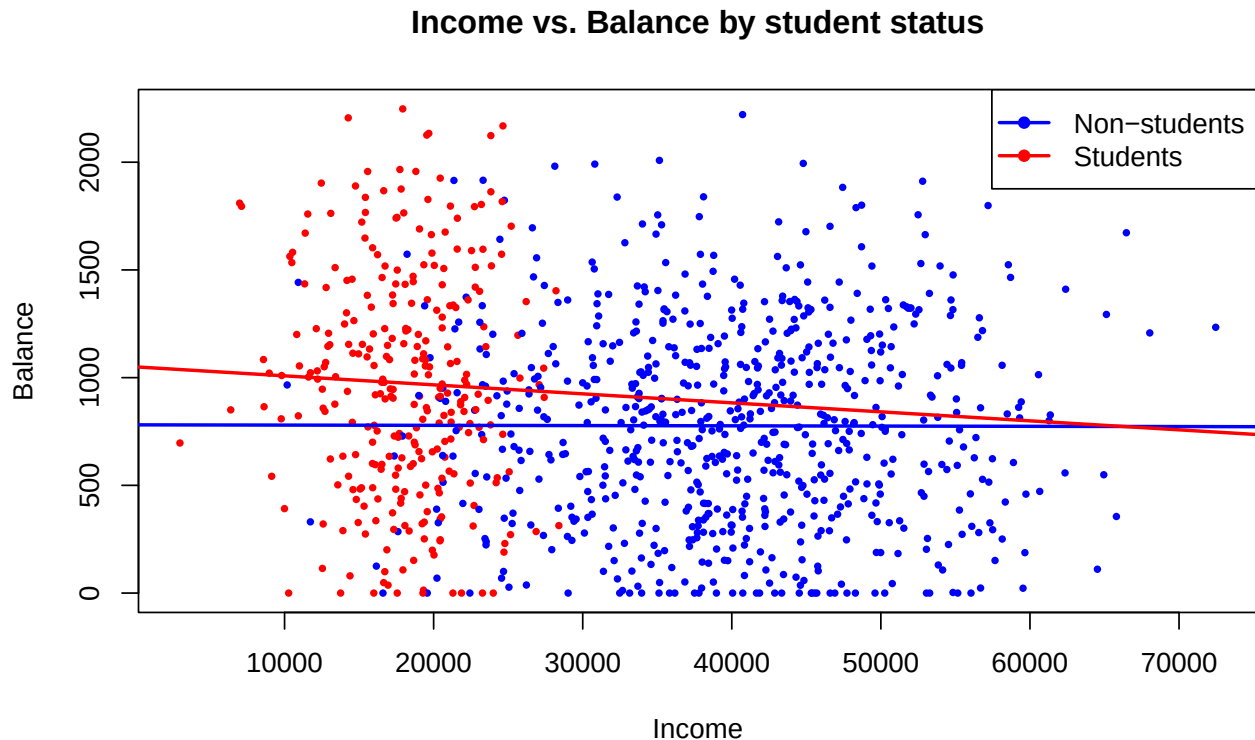
- **income:** income effect for non-students (baseline).
- **studentYes:** difference in intercept between the two groups.
- **income:studentYes:** difference in the **slope coefficient** – how strongly does the income effect differ between students and non-students?
- A **significant interaction** means: the effect of one predictor depends on the value of the other; the regression lines are **no longer parallel**.

3.6 Step 6: Visualisation of the interaction (f)

```
plot(default$income, default$balance,
     col = ifelse(default$student == "Yes", "red", "blue"),
     pch = 16, cex = 0.6,
     xlab = "Income", ylab = "Balance",
     main = "Income vs. Balance by student status")

abline(a = coef(mod_si_int)[1],
       b = coef(mod_si_int)[2],
       col = "blue", lwd = 2)
abline(a = coef(mod_si_int)[1] + coef(mod_si_int)[3],
       b = coef(mod_si_int)[2] + coef(mod_si_int)[4],
       col = "red", lwd = 2)

legend("topright", legend = c("Non-students", "Students"),
     col = c("blue", "red"), pch = 16, lwd = 2)
```



Assessment: Are the two lines roughly parallel? Then the interaction is small. If the slopes differ markedly, this argues for a genuine interaction.

3.7 Step 7: Model comparison (g)

```
anova(mod_si, mod_si_int)

## Analysis of Variance Table
##
## Model 1: balance ~ income + student
## Model 2: balance ~ income * student
##   Res.Df      RSS Df Sum of Sq    F Pr(>F)
## 1     997 232536475
## 2     996 232456061    1    80414 0.3445 0.5573
```

Interpretation: The F-test checks whether the model with interaction is significantly better than the one without. If $p > 0.05$: the added value of the interaction is not statistically supported – the simpler model is preferred.

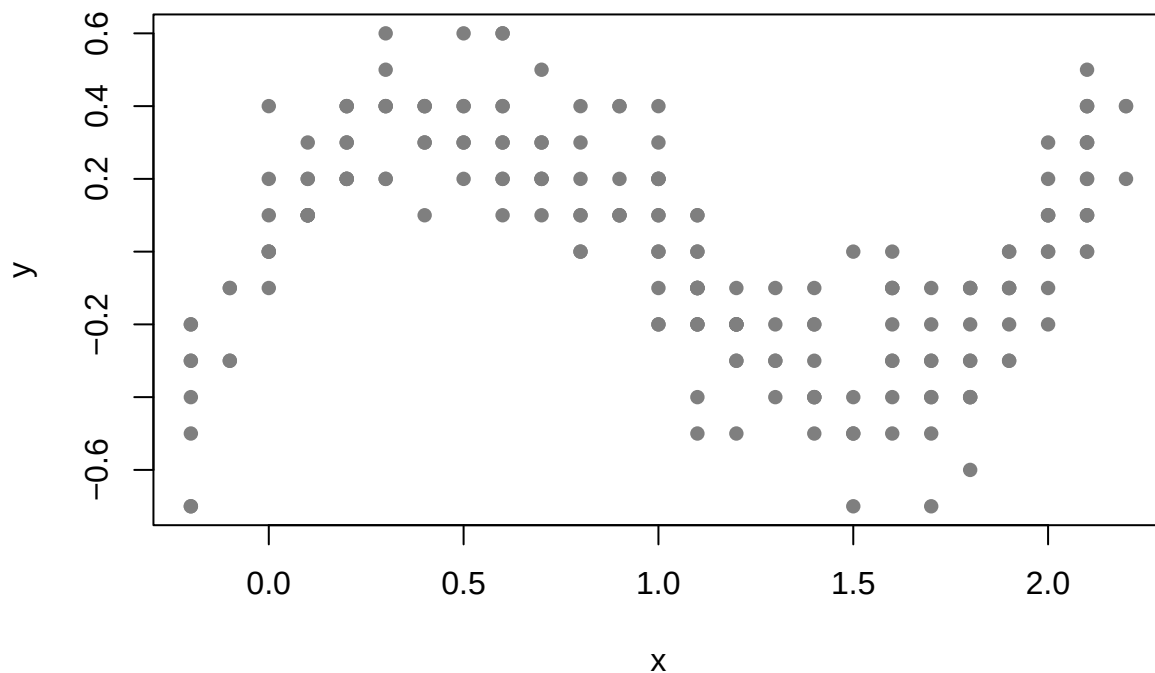
4 Exercise 3: Polynomial – Polynomial fitting

Task: For the Polynomial dataset, investigate which polynomial degree best describes the relationship between x and y .

4.1 Step 1: Read and visualise the data (a)

```
poly_dat <- read_excel("../data/WDDA_08.xlsx", sheet = "Polynomial")
plot(poly_dat$x, poly_dat$y, pch = 16, col = "gray50",
     xlab = "x", ylab = "y",
     main = "Scatterplot Polynomial")
```


Scatterplot Polynomial



Assessment: Clearly non-linear shape – a linear model will not describe the data well.

4.2 Step 2: Fit polynomials of various degrees (b)

```
m1 <- lm(y ~ x, data = poly_dat) # Linear
m2 <- lm(y ~ poly(x, 2, raw = TRUE), data = poly_dat) # Quadratic
m3 <- lm(y ~ poly(x, 3, raw = TRUE), data = poly_dat) # Cubic
m4 <- lm(y ~ poly(x, 4, raw = TRUE), data = poly_dat) # Degree 4
```

```
cat("R-squared values:\n")
```

```
## R-squared values:
```

```
cat("Degree 1: ", round(summary(m1)$r.squared, 4), "\n")
```

```
## Degree 1: 0.0525
```

```
cat("Degree 2: ", round(summary(m2)$r.squared, 4), "\n")
```

```
## Degree 2: 0.0526
```

```
cat("Degree 3: ", round(summary(m3)$r.squared, 4), "\n")
```

```
## Degree 3: 0.7286
```

```
cat("Degree 4: ", round(summary(m4)$r.squared, 4), "\n")
```

```
## Degree 4: 0.7286
```

4.3 Step 3: Model comparison with F-tests (c)

```
anova(m1, m2, m3, m4)

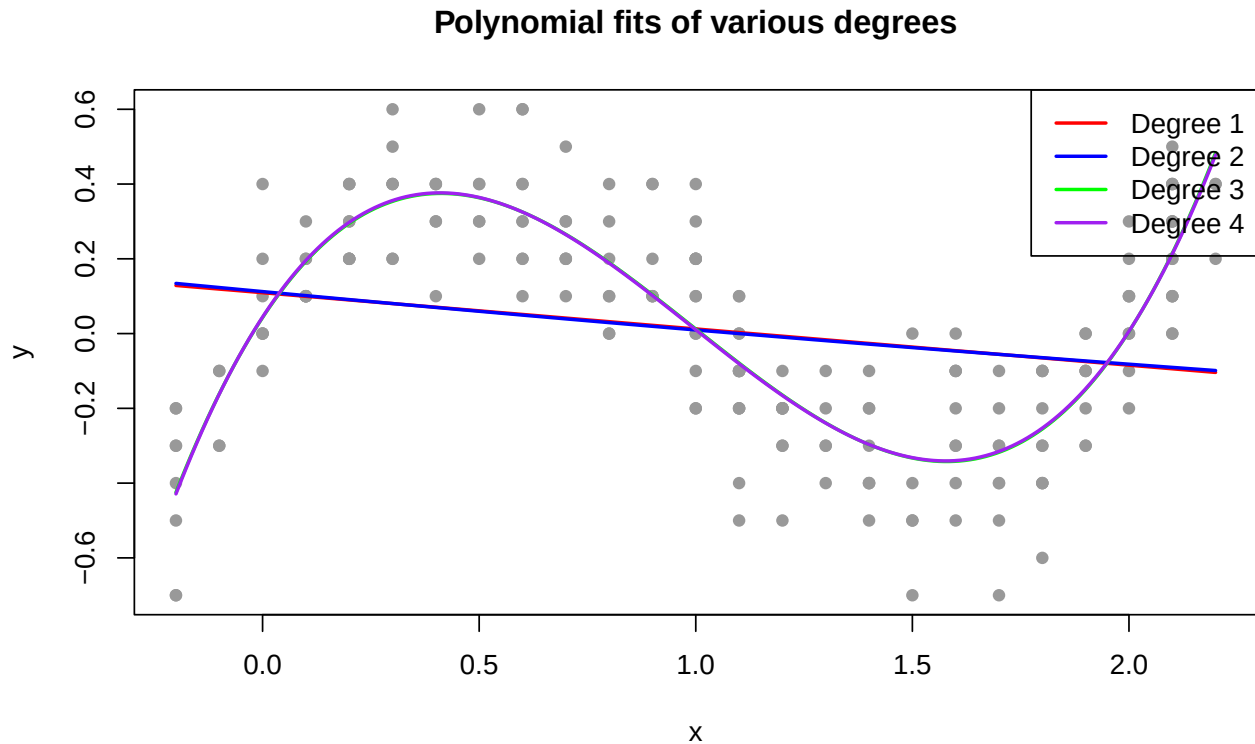
## Analysis of Variance Table
##
## Model 1: y ~ x
## Model 2: y ~ poly(x, 2, raw = TRUE)
## Model 3: y ~ poly(x, 3, raw = TRUE)
## Model 4: y ~ poly(x, 4, raw = TRUE)
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     198 16.6745
## 2     197 16.6732  1    0.0013   0.0521 0.8198
## 3     196  4.7762  1   11.8970 485.7965 <2e-16 ***
## 4     195  4.7755  1    0.0008   0.0307 0.8612
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation: The ANOVA table shows whether the next-higher polynomial degree adds significant improvement. From the degree at which $p > 0.05$, the additional complexity is no longer worthwhile.

4.4 Step 4: Visualisation of the fitted curves (d)

```
x_grid <- seq(min(poly_dat$x), max(poly_dat$x), length.out = 200)

plot(poly_dat$x, poly_dat$y, pch = 16, col = "gray60",
     xlab = "x", ylab = "y",
     main = "Polynomial fits of various degrees")
lines(x_grid, predict(m1, newdata = data.frame(x = x_grid)),
     col = "red", lwd = 2)
lines(x_grid, predict(m2, newdata = data.frame(x = x_grid)),
     col = "blue", lwd = 2)
lines(x_grid, predict(m3, newdata = data.frame(x = x_grid)),
     col = "green", lwd = 2)
lines(x_grid, predict(m4, newdata = data.frame(x = x_grid)),
     col = "purple", lwd = 2)
legend("topright", legend = c("Degree 1", "Degree 2", "Degree 3", "Degree 4"),
     col = c("red", "blue", "green", "purple"), lwd = 2)
```



Assessment: Higher degrees follow the data more closely, but beware: too high a polynomial degree leads to **overfitting** – the model adapts to random fluctuations without better describing the true relationship.

5 Summary

This problem set has shown:

1. **Variable transformations** (square-root, logarithm, polynomials) extend the linear model to non-linear relationships.
2. R^2 **comparisons** are only meaningful when the response variable is the same (caution with log transformations of the response).
3. **Residual analyses** are essential for assessing whether a model fits the data well – patterns in the residuals reveal model misspecification.
4. **Categorical predictors** are coded via dummy variables. The choice of baseline affects the interpretation of the coefficients but not the predictions.
5. **Interactions** allow the effect of one predictor to depend on the value of another – geometrically this corresponds to non-parallel regression lines.
6. **F-tests** for model comparison help to choose the right complexity – simpler is better unless the simpler model is significantly worse.